MIL-A-21577/6A(SHIPS)
28 September 1962
SUPERSEDING
MIL-A-21577/6(SHIPS)
15 September 1959

MILITARY SPECIFICATION

AMPLIFIER ASSEMBLY, AM-2316/SIA

1. SCOPE

1.1 This specification covers the detailed requirements for an audio amplifier control assembly, AM-2316/SIA.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-A-21577 - Amplifier, Audio Frequency; and Amplifier-Control Groups - Shipboard Announcing.

STANDARDS

MILITARY
MIL-STD-200 - Electron Tubes, Selection and Use of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Amplifier Assembly AM-2316/SIA shall be in accordance with all applicable paragraphs of MIL-A-21577 and as specified herein. Whenever a requirement of MIL-A-21577 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.2 Equipment complement shall consist of two power amplifiers in a common deck mounted enclosure. Maximum dimensions shall not exceed the following: Height 60 inches; width 32 inches; and depth 20 inches.

3.3 Wiring.- Wiring employed shall have insulation type T as shown in the applicable table of MIL-I-983.

3.4 Electron tubes.- Electron tubes shall be selected from MIL-STD-200.

3.5 Forced air cooling.- In addition to the requirements specified in MIL-A-21577, facilities for forced air cooling will be required for power amplifiers. Blower motors shall be individually fused. Temperature requirements of the amplifier assembly shall be met with blowers in nonoperating condition.

3.6 Test facilities.- A test meter and switching facilities shall be included in each power amplifier, and shall provide a ready means for indicating the operating conditions of each vacuum tube stage, and the output signal level. The meter calibration and circuit component values shall be such that the operating condition of the amplifiers can be determined directly from the meter without reference to tables or charts.

3.7 Output power.- The rated output power for each of the two power amplifier assemblies shall be 500 watts, minimum rating.

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3.8 Interstage input circuit, input impedance, and input level will not be required.

3.9 Oscillator assemblies and preamplifier assemblies are not required.

3.10 Input facilities.- Amplifier Assembly AM-2316/SIA shall have the following input facilities:

3.10.1 Terminals shall be provided for input from preamplifier assembly to each power amplifier.

3.10.2 Terminals shall be provided for input controls from Amplifier-Control Assembly AN/SIA-117 or AN/SIA-118.

3.11 Input impedance.- The input impedance of the power amplifier assembly shall be 8000 ohms (bridging).

3.12 Output facilities.- Amplifier Assembly AM-2316/SIA shall have the following output facilities:

3.12.1 Output voltage.- Each amplifier assembly shall have facilities which will provide either 70 ± 4 volts root mean square (r.m.s.) at rated output power, or 95 ± 5 volts r.m.s. at rated output power.

3.12.2 Terminals shall be provided for 70-volt, 95-volt, and low output from each power amplifier.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling, examination, and tests shall be as specified in MIL-A-21577.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-A-21577.

6. NOTES

6.1 The notes specified in MIL-A-21577 are applicable to this specification.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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